Lesson 2: Key Press Reactor

001 Game Creator (Windows only)

VIDEO GAME CREATION

This week: sequences, events, and the game loop. Your job is to build something that reacts when you press a key - and this is also the start of your Semester 1 project, so save it somewhere safe and give it a name you'll recognize.

Tip: this lesson's Assets folder already has a ready-to-use placeholder character image (and an optional sound effect for Level 3) - you don't need to find or make your own art today.

Setup

1. Create a new project named [YourName]Game from the Blank Template - a Warning will note this template is for experienced users with no pre-built input, menus, or actors; that's fine, click Continue. Blank starts with no Map and nothing placed on screen, but it still includes one default Player/Party roster entry named "Hero" - it just has no appearance or map location yet. The next four steps get Hero on screen.
2. Create a Map: click the Maps tab in the small tabbed panel at the bottom-left, then click New. In the Map's Properties, set General > View to Top (top-down, matching our other engines), then give the Map a name (e.g. Level1).
3. Give Hero an appearance: go to Resources > Sprites > Create New > No Template, then under the Graphics section use "Add Animation > Image Files" to import one of the placeholder images from this lesson's Assets folder - Resources > Import Resources does NOT accept plain image files, only pre-formatted asset sheets, so the Sprite Editor is the correct path for a simple placeholder PNG. In the Sprite Editor, check the "Face X" box just left of the preview image, then click the "Top" tab above the preview to confirm it looks right from a top-down view.
4. Attach the sprite: go to Resources > Players and Party Members > Hero > Edit Actor > Sprite section > Body, and choose the Sprite you just created.
5. Set the starting point: open Game Settings from the top menu, set Starting Map to the Map from step 2, then set Starting Location by dragging Hero's icon onto the map in the popup window.
6. Press F5 to test - Hero's sprite should appear at the spot you set. Once you see it, you're ready for Level 1 below.

Level 1 - Minimal (MVP)

One key press causes one visible reaction. This is a complete finish for today - reaching this is enough to move on to next week.

7. Go to Resources > Input Triggers.

8. Click "Add Control Combination," press the Right Arrow key, then click "Pressed."
9. Click "Edit Script" for that trigger and add a Position Actor event (or Take Movement in Direction for tile-based movement) targeting main, moving it right a small amount.
10. Save, then click Play/Test to try it.

Alternate Path: Textual Scripting (typed BASIC-like code)

001 Game Creator has a second scripting mode that uses typed, BASIC-like commands instead of the graphical event blocks you used above.

IMPORTANT - READ BEFORE YOU TRY THIS: switching a project's Mode from Graphical to Textual is PERMANENT once you save. Do NOT convert your main [YourName]Game project. Create a separate throwaway test project for this exercise only.

11. Create a new, separate project (e.g. [YourName]GameTextual) - do not use your main Semester 1 project for this.
12. Open the Mode menu and switch this new project to Textual (confirm with your instructor that you understand this cannot be undone once saved).
13. Open the Script Editor for a Right Arrow Input Trigger, same as before.
14. Instead of dragging graphical event blocks, type the command directly - start typing "Position" and use the editor's autocomplete/command list to find the text equivalent of the Position Actor event you used in the graphical path, targeting main with a relative X offset.
15. Save and Test to confirm the same result as your graphical version.

Exact command names can vary slightly by version - the Script Editor's autocomplete and the 001gamecreator.com/wiki Textual Scripting reference show the current syntax. If you want to keep exploring this mode, do it in your throwaway test project, not your real one. Level 3's five goals (animation, sound, recolor, clamp, and grouping it all under each direction's script) all have Textual equivalents too - same ideas as the graphical path, just typed instead of dragged.

Level 2 - Complete (Alpha/Beta)

All 4 arrow keys move your character in 4 directions.

- Repeat for Left, Up, and Down arrow keys as three more Control Combinations, each moving main in that direction.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Animate: since movement here is press-and-release rather than press-and-hold, every key release is a real (if brief) idle moment, even mid-walk - so the idle-reset script you build for the Recolor goal below already makes main blink between its direction color and idle color on every step, with no separate event needed. See Recolor for the how.
- Sound: play sfx_blip.wav (Speaker event) each time a movement key is pressed - naturally silent whenever no key is being pressed, since Input Triggers fire on press, not continuously.
- Recolor: pair each direction's Position Actor event with a Change Body/Clothing Sprite event, swapping main's whole Sprite for a different pre-made colored one rather than tinting the sprite it already has - import the other placeholder circles via Resources > Sprites the same way you set up main's original appearance. To reset the sprite when idle, add a second "Released" Control Combination for each arrow key, and use a Multi-

Control Branch checking all 4 arrow keys - its "none pressed" path is the real idle moment, so put the reset-to-default sprite swap there.

- Clamp: nothing to build - 001 Game Creator does this one automatically. Open your Map's Properties (double-click the Map, or Map menu > Properties) and look at the Boundaries section: all 4 directions default to ticked with no Map set in their Connection dropdowns, which means the engine itself blocks main from ever crossing an edge. See the Key for what would happen if you changed that.
- Structure: keep each direction's movement, sound, and recolor events together in that direction's own Input Trigger script - not scattered into separate triggers.

If Something's Wrong

- Nothing happens on a key press: confirm the Input Trigger is actually set to "Pressed," and that the script attached to it is connected to the Start node in the Script Editor.
- Dialog fields may look slightly different depending on your version - the Script Editor and Actor pages at 001gamecreator.com/wiki have the current layout.

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001gamecreator.com/wiki, Student Resource List > 001 Game Creator